

1 Choose the correct statement regarding fructose-1,6-bisphosphatase

- a. It catalyzes glycolytic reaction
- b. Its expression is induced during fasting
- c. Its activity is increased by fructose-2,6-bisphosphate (F-2,6-BP)
- d. Its activity is increased by AMP
- e. It catalyzes reversible reaction in gluconeogenesis

Tetracycline blocks protein synthesis by:

- a. Inhibiting peptidyl transferase
- b. Inhibiting initiation of translation
- c. Inhibiting termination of translation
- d. Inhibiting binding of tRNA with ribosome
- e. Inhibiting translocase enzyme

Which of the following statements about creatinine measurement is false?

- a. The initial absorbance for the standard and test is recorded exactly after 30 seconds
- b. Creatinine under alkaline conditions reacts with picrate forming a red-colored complex
- c. Decrease in absorbance measured at 510 nm
- d. Serum creatinine level is affected by muscle mass
- e. Increase in absorbance measured at 540 nm

One of the following enzymes is not of intestinal origin?

- a. Lactase
- b. Isomaltase
- c. Maltase
- d. α -amylase
- e. Sucrase

5 Among the following, one of the amino acid catabolism must involve extrahepatic tissues for transamination.

- a. Tryptophan
- b. Isoleucine
- c. Threonine
- d. Aspartate
- e. Glutamine

The region of DNA known as TATA BOX is the site for binding of:

- a. DNA topoisomerase
- b. Peptidyl transferase
- c. DNA-dependent RNA polymerase
- d. Polynucleotide phosphorylase
- e. DNA polymerase

Which condition is associated with a positive nitrogen balance?

- a. Trauma
- b. Uncontrolled DM
- c. Cancer
- d. Starvation
- e. Pregnancy

Intestinal nucleosidases act on nucleosides and produce:

- a. Purine base only
- b. Phosphate only
- c. Purine and pyrimidine base and phosphate
- d. Purine or pyrimidine bases and sugars
- e. Sugar only

9 The fatty acid synthase complex of mammals

- a. Catalyzes eight different enzymatic steps
- b. Is a dimer of different subunits
- c. Is composed of four different proteins
- d. Is composed of seven covalently linked enzymes
- e. Dissociates into eight different proteins

Which of the following is correct statement about payoff phase of glycolysis?

- a. Produces pyruvate
- b. Consumes ATP
- c. Does not involve oxidation-reduction reaction
- d. Consumes H₂O
- e. Inorganic phosphate is released

Which one of the following is carried out by ribozyme property of large subunit of ribosome?

- a. Termination of translation
- b. Peptidyl transferase activity
- c. Complimentary attachment of anticodon to codon
- d. Amino acid activation
- e. Growth of the peptide chain

The mushroom poison amanitin is an inhibitor of:

- a. Purine nucleotide synthesis
- b. DNA synthesis
- c. Protein synthesis
- d. Pyrimidine nucleotide
- e. mRNA synthesis

13 A variable that has NO EFFECT on reaction rate is:

- ☒ a. none of the above
- ☐ b. catalyst
- ☐ c. concentration
- ☐ d. energy of activation
- ☐ e. temperature

Which is the most abundant type of RNA?

- ☒ a. mRNA
- ☐ b. rRNA
- ☐ c. pre-mRNA
- ☐ d. tRNA
- ☐ e. hnRNA

Which of the following have an autocatalytic activity?

- ☒ a. Carboxypeptidases
- ☐ b. Dipeptidase
- ☐ c. Tripeptidase
- ☐ d. Elastases
- ☐ e. Trypsin

Which of the following statements is correct?

- ☒ a. Cyanide, oligomycin, and 2,4-dinitrophenol compete with O_2 for cytochrome oxidase (Complex IV).
- ☐ b. Oligomycin and cyanide inhibit synthesis of ATP; 2,4-dinitrophenol inhibits the respiratory chain.
- ☐ c. Cyanide and 2,4-dinitrophenol inhibit the respiratory chain, and oligomycin inhibits the synthesis of ATP.
- ☐ d. Cyanide inhibits the respiratory chain, whereas oligomycin and 2,4-dinitrophenol inhibit the synthesis of ATP.
- ☐ e. Oligomycin inhibits the respiratory chain, whereas cyanide and 2,4-dinitrophenol prevent the synthesis of ATP.

17 The 16S r RNA of smaller prokaryotic ribosomal subunit has complementary sequence with that of the mRNA, called:

- a. Hogness sequence
- b. GAAT box
- c. TATA box
- d. Pribnow sequence
- e. Shine-Dalgarno sequence

Choose the correct statement regarding regulation of glycolysis in muscle.

- a. It is inhibited by high levels of fructose-2,6-bisphosphate
- b. It is inhibited by high levels of ADP
- c. It is activated by high levels of citrate
- d. It is activated by high level of ATP
- e. It is activated by high levels of AMP

How many codons can be translated to specific amino acids?

- a. 25
- b. 61
- c. 64
- d. 48
- e. 45

Certain pathways, such as those involved in the disposal of toxic compounds, are feed-forward regulated, in which regulation may not occur through

- a. increasing the product concentration for feedback inhibition
- b. allosteric activation of a rate limiting enzyme through a compound related to substrate supply,
- c. an increased supply of substrate to an enzyme with a high K_m ,
- d. increased concentration of a hormone that stimulates a storage pathway by controlling enzyme phosphorylation state
- e. substrate-related induction of gene transcription

21 Complete oxidation of glucose into CO_2 and H_2O occurs in all of the following cells except

- a. Liver
- b. Kidney cells
- c. Red blood cells
- d. Muscle
- e. Brain

The urine of patients with Alkaptonuria turns dark when exposed to light and air because of

- a. Oxidation & polymerization of homogentisate
- b. Deamination of phenylalanine to phenylpyruvate
- c. Polymerization of tyrosine & homogentisate
- d. Oxidation of fumarylacetoacetate
- e. Polymerization of phenylalanine

Which of the following compounds is detoxified by Glucuronidation reactions?

- a. Phenylacetate
- b. Aspirin
- c. Methanol
- d. Bromobenzene
- e. Bilirubin

Which one of the following point mutation type produces a normally functioning protein

- a. insertion mutation
- b. non-sense mutation
- c. silent mutation
- d. missense mutation
- e. frame-shift mutation

25 The most efficient substrate of an enzyme is usually considered to be the substrate with the

- a. largest K_m
- b. smallest k_{cat}/K_m
- c. smallest K_m
- d. largest k_{cat}
- e. largest k_{cat}/K_m

The active molecule that is stabilized by a vasodilator and neurotransmitter nitric oxide radical is

- a. Histidine decarboxylase
- b. Dopamine
- c. Phosphodiesterase
- d. Cyclic GMP
- e. Guanosine monophosphate

When the liver is actively synthesizing fatty acids, a concomitant decrease in β -oxidation of fatty acids is due to

- a. Decrease in adipocyte lipolysis
- b. Activation of an enzyme
- c. Inhibition by an end product
- d. Detergent effects
- e. Inhibition of translocation between cellular compartments

If the mitochondria were blocked at the site of NADH oxidation and were treated with succinate as substrate, what would be the P:O ratio?

- a. Same as that normally produced by succinate
- b. One less than normally produced by succinate
- c. Zero
- d. One more than normally produced by succinate
- e. Higher than normal because of the excessive heat produced from uncoupling

- 29 The equation below refers to the reaction: $A + B \rightleftharpoons C + D$, which $\Delta G = \Delta G^\circ + RT \log_e \left\{ \frac{[C][D]}{[A][B]} \right\}$. At the starting concentrations of $[A]_0$, $[B]_0$, $[C]_0$, and $[D]_0$, no change occurs upon addition of enzyme. You may, however, force the reaction to proceed to the right by
- a. adding more A, B, C, and D to increase their concentration by the same factor.
 - b. adding buffer so as to reduce the concentrations of A, B, C, and D by the same factor.
 - c. increasing $[C]$ or $[D]$
 - d. increasing $[A]$ or $[B]$
 - e. decreasing $[A]$ or $[B]$.

Which of the following is true statement about the Cori cycle?

- a. It is a means of sustaining energy source for aerobically exercising muscle
- b. Muscle produces lactate that ultimately reaches the liver
- c. It is a means of generating energy source for liver
- d. Muscle forms glucose from the precursor molecules
- e. Liver metabolizes glucose to generate pyruvate

The number of base pair in a single turn of B-form DNA about the axis of the molecule is

- a. 4
- b. 12
- c. 6
- d. 10
- e. 8

Which of these statements about enzyme-catalyzed reactions that obey Michaelis-Menten kinetics is false?

- a. The activation energy for the catalyzed reaction is the same as for the uncatalyzed reaction
- b. At saturating levels of substrate, the rate of an enzyme-catalyzed reaction is proportional to the enzyme concentration.
- c. If enough substrate is added, the normal V_{max} of a reaction can be attained even in the presence of a competitive inhibitor.
- d. The rate of a reaction decreases steadily with time as substrate is depleted.
- e. The Michaelis-Menten constant K_m equals the $[S]$ at which $V = 1/2 V_{max}$.

33 The action of T-amylase on starch produces all of the following except

- a. D-dextrins
- b. Trisaccharides
- c. Isomaltose
- d. Trehalose
- e. Maltose

The advantage of fat as an energy source is that

- a. It is more calorific on a per gram basis than other foodstuffs
- b. It needs less oxygen than other foodstuffs for complete oxidation
- c. It is more readily soluble than other foodstuffs
- d. It needs high oxygen for complete oxidation
- e. It is more chemically reactive than other foodstuffs

During replication process deoxyribonucleotides are added to RNA primer by:

- a. DNA polymerase II
- b. DNA polymerase I
- c. σ -Subunit of DNA polymerase II
- d. DNA polymerase III
- e. DNA polymerases I & II

A particular triplet of bases in the coding sequence of DNA is AAA. The anticodon on the tRNA that binds the mRNA codon is:

- a. either UAA or TAA, depending on first base wobble
- b. UUA
- c. UUU
- d. AAA
- e. TTT

37 Which of the following is not true about pentose phosphate pathway?

- a. It is a means for catabolism of excess pentoses
- b. It provides a means for interconversion of pentoses
- c. It provides precursor for nucleotide synthesis
- d. It is the primary means of ATP formation in cells lacking mitochondria
- e. It provides reductant molecule

One of the following statements concerning fatty acid structure and its oxidation is CORRECT?

- a. They are called monounsaturated when they contain more than two carbon-carbon double bonds.
- b. They consist of a carboxylic acid head group attached to a carbohydrate chain.
- c. Their melting points increase with increasing unsaturation.
- d. During the β -oxidation of fatty acid, trans double bond is formed by Acyl CoA dehydrogenase.
- e. They occur in the body mainly in the form of free (non-esterified) fatty acids.

The correct statement concerning RNA and DNA polymerases is:

- a. Both polymerases can add nucleotides at 3' and 5' end of the growing chain
- b. RNA polymerase use nucleoside diphosphates as a substrate
- c. DNA polymerases can add nucleotides at both ends of the chain
- d. RNA polymerase require primers and add bases at 5' end of the growing chain
- e. Both polymerases can add nucleotides only at the 3' end of the growing chain

α -Linolenic acid is considered to be nutritionally essential in humans BECAUSE:

- a. It contains three double bonds.
- b. In humans double bonds cannot be introduced into fatty acids beyond the $\Delta 9$ position.
- c. It is an $\omega 3$ fatty acid.
- d. In humans double bonds cannot be introduced into fatty acids beyond the $\Delta 12$ position.
- e. Human tissues are unable to introduce a double bond in the $\Delta 9$ position of fatty acids.

- 41 During oxidative phosphorylation, the proton motive force that is generated by electron transport is used to:
- a. induce a conformational change in the ATP synthase.
 - b. reduce O_2 to H_2O .
 - c. generate the substrates (ADP and P_i) for the ATP synthase.
 - d. oxidize NADH to NAD^+ .
 - e. create a pore in the inner mitochondrial membrane.

Which blood glucose determination method is used to evaluate the past status of blood glucose level?

- a. Peroxidase method
- b. HbA1c (glycated hemoglobin)
- c. GOD method
- d. POD method
- e. GOD-POD method

eRF1 is the release factor in eukaryotes that requires:

- a. ATP for its binding with ribosome
- b. Mn^{2+} to its binding with ribosome
- c. ATP and GTP for its binding with ribosome
- d. CTP for its binding with ribosome
- e. GTP for its binding with ribosome

What is the correct sequence of enzymes involved in the determination of total cholesterol in the blood?

- a. Cholesterol Oxidase, Cholesterol Esterase, Peroxidase
- b. Glycerol dehydrogenase, Cholesterol Oxidase, Peroxidase
- c. Cholesterol Esterase, Cholesterol Oxidase, Peroxidase
- d. Cholesterol Oxidase, Peroxidase, Cholesterol Esterase
- e. Cholesterol Esterase, Peroxidase, Cholesterol Oxidase

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The first reaction unique to purine nucleotide synthesis is catalyzed by:

- a. Formyl transferase
- b. IMP reductase
- c. PRPP glutamyl amido transferase
- d. PRPP synthetase
- e. Phosphoribosyl glycinamide synthetase

mRNA is complementary to the nucleotide sequence of:

- a. Ribosomal RNA
- b. Template strand
- c. Coding strand
- d. tRNA
- e. Sense strand

Which of the following glucose transporters serves as glucose sensor in the pancreas?

- a. GLUT3
- b. GLUT5
- c. GLUT1
- d. GLUT2
- e. GLUT4

Which of the following has the highest redox potential in the respiratory chain?

- a. Ubiquinone
- b. NAD
- c. FMN
- d. FAD
- e. O₂

49 Why are individuals deficient in glucose-6-phosphate dehydrogenase prone to hemolysis?

- a. Increased ATP level
- b. Lack of ribose-5-phosphate
- c. Lack of ribulose-5-phosphate
- d. Accumulation of NADPH
- e. Increased oxidative stress

Which one of the following apolipoproteins is synthesized in the liver as part of the coat of very-low-density lipoproteins (VLDLs)?

- a. Apo A1
- b. Apo B100
- c. Apo CII
- d. Apo B48
- e. Apo E

Which of the following enzymes controls substrate availability to β -oxidation pathway?

- a. Hormone sensitive lipase.
- b. Acyl CoA carboxylase.
- c. Acyl CoA Dehydrogenase
- d. Lipoprotein lipase.
- e. Acyl CoA-Carnitine transferase.

Which of the following is true for both prokaryotic and eukaryotic gene expression?

- a. The mRNA transcript is the exact complement of the gene from which it was copied.
- b. After transcription, a 3' poly-A tail and a 5' cap are added to mRNA.
- c. mRNA is synthesized in the 3' $\rightarrow$ 5' direction.
- d. RNA polymerase binds to the promoter region to begin transcription.
- e. Translation of mRNA can begin before transcription is complete.

53 Which one is an oxidation-reduction coenzyme?

- a. Biotin
- b. FAD
- c. Pyridoxal phosphate (PLP)
- d. CoASH
- e. Thiamine pyrophosphate (TPP)

Which one of the following does not require primer in order to function?

- a. DNA dependent DNA pol I
- b. RNA dependent DNA pol II
- c. DNA dependent RNA pol
- d. RNA dependent RNA pol
- e. RNA dependent DNA pol III

Which one of the following energy rich molecules is required for initiation of translation?

- a. ATP
- b. CTP
- c. TTP
- d. AMP
- e. GTP

In a DNA molecule the thymine concentration is 30%, the guanosine concentration will be:

- a. 10%
- b. 40%
- c. 20%
- d. 25%
- e. 30%

CCAAT box is present in

- a. Prokaryotic promoters 10 bp upstream of transcription start site
- b. Eukaryotic promoters 25 bp upstream of transcription start site
- c. 10 bp downstream of the transcription start site of prokaryotes
- d. Prokaryotic promoters 35 bp upstream of transcription start site
- e. Eukaryotic promoters 70–80 bp upstream of transcription start site

Polysomes are:

- a. Aggregation of many mRNA with a single ribosome during translation
- b. Aggregation of mRNA molecules during replication
- c. Aggregation of many ribosomes with a single mRNA during translation
- d. Aggregation of many ribosomes during transcription
- e. Aggregation of elongation factors with mRNA during transcription

Degeneracy of genetic code implies that:

- a. multiple codons decode the same amino acid
- b. no anticodon on tRNA molecule
- c. codons do not code for specific amino acid
- d. specific codon decodes many amino acids
- e. unavailability of codon for anticodon

Which of the following process is not involved in the post transcriptional process of mRNA?

- a. Base modification
- b. Removal of introns
- c. 5' capping of mRNA
- d. Splicing
- e. Addition of poly A tail

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Which one is a physically separated but physiologically-coupled process?

- a. Deamidation
- b. Hydrolytic deamination
- c. Oxidative deamination
- d. Transdeamination
- e. Non-oxidative deamination

Which of the following kinetic parameters remains the same for substrate to change to P, whether the reaction is enzyme-catalyzed or uncatalyzed?

- a. V_0
- b. V
- c. k
- d. K_{eq}
- e. K_m

Which of the following is correct about glycogen?

- a. It is composed of α -glucose units
- b. It is synthesized during fasting state
- c. It is found in significant amounts in most cells of our body
- d. Its synthesis is facilitated by glucagon
- e. It is a mobilizable form of energy storage

Which coenzyme is required for the synthesis of TMP by de novo pathway:

- a. N⁵- Formimino tetrahydrofolate
- b. N⁵, N¹⁰- Methylene tetrahydrofolate
- c. N¹⁰- ethyl tetrahydrofolate
- d. N⁵- Methyl tetrahydrofolate
- e. N¹⁰- Formyl tetrahydrofolate

65 In inherited deficiency of hypoxanthine guanine phosphoribosyl transferase:

- a. Salvage of purines is increased
- b. Salvage of purines is decreased
- c. Synthesis of uric acid is decreased
- d. Primary gout might be occurred
- e. De novo synthesis of purine nucleotides is decreased

What is a ribozyme?

- a. An enzyme that synthesizes RNA as part of the transcription process
- b. An enzyme that uses RNA as a substrate
- c. An enzyme that catalyzes the association between the large and small ribosomal subunits
- d. An enzyme that synthesizes RNA primers during DNA replication
- e. An RNA with enzymatic activity

The removal of free ammonia from the side chain of glutamine requires

- a. L-Amino Acid oxidase
- b. D-amino acid oxidase
- c. Aminotransferase
- d. Glutaminase
- e. Glutamate Dehydrogenase

Non-coding sequences in a gene are known as:

- a. Missense codons
- b. Cistrons
- c. Introns
- d. Exons
- e. Nonsense codons

69 The most common transport method for absorption of L-amino acids from small intestine after digestion is

- a. H⁺-driven primary active transport
- b. Peptide carrier-mediated secondary active transport
- c. Na⁺-driven secondary active transport
- d. Carrier-mediated passive diffusion
- e. Primary active Na⁺-amino acid antiport

The end product of cytosol fatty acid synthetase in humans is

- a. Arachidonic acid
- b. Palmitic acid
- c. Palmitoleic acid
- d. Oleic acid
- e. Linoleic acid

Which of the following fasting blood glucose reference range shows diabetic condition? (Considering the test was done on more than one occasion)

- a. ≥ 200 mg/dl
- b. 100-125 mg/dl
- c. ≥ 126 mg/dl
- d. 140-199 mg/dl
- e. 200-300 mg/dl

Reduction of ribonucleotide diphosphates (NDPs) to their corresponding deoxy ribonucleotide diphosphates (dNDPs) involves:

- a. NAD
- b. FAD
- c. TPP
- d. NADPH
- e. FMN

73 A transcription unit that is 8,000 nucleotides long may use 1,200 nucleotides to make a protein consisting of approximately 400 amino acids. This is best explained by the fact that:

- a. there are termination exons near the beginning of mRNA.
- b. there is redundancy and ambiguity in the genetic code.
- c. nucleotides break off and are lost during the transcription process.
- d. many nucleotides are needed to code for each amino acid.
- e. many noncoding stretches of nucleotides are present in mRNA.

In biosynthesis of proteins the chain terminating codons are:

- a. GCG, GCA and GCU
- b. AAU, AAG and GAU
- c. TAG, GAG and UGG
- d. UGG, UGU and AGU
- e. UAA, UAG and UGA

One of the following statements is correct concerning the conversion of pyruvate to acetyl-CoA

- a. Inhibition of the reaction facilitates lactate formation
- b. Occurs in all cells of our body
- c. Vitamin deficiency does not affect the reaction
- d. It is catalyzed by monomeric enzyme
- e. Can occur under aerobic and anaerobic conditions

Choose the correct statement about glycogenesis

- a. It does not require energy investment
- b. It mainly occurs in liver and muscle
- c. Free glucose is a substrate for glycosidic bond formation
- d. Doesn't need a primer
- e. It involves formation of 1-1,4-glycosidic linkages

77 Although each specific tRNA differs from the others in its sequence of nucleotides, all tRNA molecules contain a base paired stem that terminates in the sequence CCA at:

- a. Additional arm termini
- b. D-termini
- c. 3' termini
- d. Anticodon arm
- e. 5' termini

During DNA replication, the unwound strands of DNA are held apart by:

- a. Helicase
- b. Ligase
- c. Double strand binding protein
- d. Topoisomerase
- e. Single strand binding protein

Translocation of the newly formed peptidyl tRNA at the A site into the empty P site involves:

- a. EF-I, GDP
- b. EF-II, GTP
- c. EF-I, GTP
- d. EF-III, GTP
- e. Peptidyl transferase, GTP

Which one of the following proteins can identify termination signal for the process of transcription?

- a. Rho factor
- b. Sigma factor
- c. Epsilon factor
- d. Alpha factor
- e. Psi factor

81 All of the following statements about nonsense codons are correct except:

- a. They act as chain termination signals
- b. They are identical in nuclear and mitochondrial DNA
- c. They have complementary anticodons
- d. They can code for amino acids
- e. None of the above

Which of the following is INCORRECT about the Cytochrome P 450 enzyme system?

- a. All of them are hemoproteins
- b. They are dioxygenases in their mechanism of action
- c. Some exhibit genetic polymorphism
- d. They can be inhibited by drugs
- e. Most of these enzymes are inducible enzymes

The size of tRNA ranges from:

- a. 50–100 nucleotides
- b. More than 320 nucleotides
- c. 90–300 nucleotides
- d. 20–30 nucleotides
- e. 0–40 nucleotides

Which of the following steps in the biosynthesis of cholesterol is thought to be rate-controlling and the locus of metabolic regulation?

- a. Squalene →lanosterol
- b. Lanosterol →cholesterol
- c. Geranyl pyrophosphate →farnesyl pyrophosphate
- d. Mevalonic acid →geranyl pyrophosphate
- e. 3-hydroxy-3-methylglutaryl CoA →mevalonic acid

85 Erythromycin acts on ribosomes and inhibit:

- a. Binding of mRNA
- b. Translocation
- c. Peptidyl transferase activity
- d. Formation of initiation complex
- e. Binding of aminoacyl tRNA

Conversion of inosine monophosphate to xanthine monophosphate is catalyzed by:

- a. Formyl transferase
- b. Xanthine-guanine phosphoribosyl transferase
- c. HGPRT
- d. Adenine phosphoribosyl transferase
- e. IMP dehydrogenase

Among the following mechanisms of amino acid synthesis, which type of reaction does not involve transamination reaction?

- a. Aspartate from oxaloacetate
- b. Serine from 3-phosphoglycerate
- c. Alanine from pyruvate
- d. Tyrosine from phenylalanine
- e. Glutamate from α -ketoglutarate

During prokaryotic transcription, the promoter region is recognized by:

- a. γ subunit of RNA polymerase
- b. σ factor of RNA polymerase
- c. DNA binding proteins
- d. β subunit of RNA polymerase
- e. Rho factor

89 Choose the incorrect statement concerning glycogen degradation in liver.

- a. It produces glucose-6-phosphate that is oxidized by liver itself
- b. It occurs during fasting state
- c. It provides glucose that is metabolized by brain
- d. It provides glucose that maintains normal glucose level
- e. The process is facilitated by glucagon

Which of the organs listed below is incorrectly matched with a substance it normally stores (S) or uses as fuel (F)?

- a. Muscle: glycogen (S)
- b. Heart: fatty acids (F)
- c. Adipose tissue: fatty acids (F)
- d. Liver: fatty acids (F)
- e. Red blood cell: ketone bodies (F)

Simultaneous fatty acid oxidation and synthesis is prevented by which of the following molecule?

- a. Acetyl CoA
- b. Palmitate
- c. β -keto-acyl CoA dehydrogenase
- d. Thiolase
- e. Malonyl CoA

Transcription in eukaryotes requires which of the following in addition to RNA polymerase?

- a. aminoacyl synthetase
- b. several transcription factors (TFs)
- c. start and stop codons
- d. ribosomes and tRNA
- e. the protein product of the promoter

93 During fatty acid metabolism in humans, coenzyme A (CoA) is different from acyl carrier protein (ACP) in which one of the following ways?

- a. Binding of fatty acids
- b. Function in the cytosol
- c. Function in fatty acid oxidation
- d. Binding of malonic acid with a phosphopantetheine
- e. Function in fatty acid synthesis

All of the following are the major detoxification reaction in phase-1, EXCEPT?

- a. Acetylation
- b. Epoxidation
- c. Hydrolysis
- d. Reduction
- e. Oxidation

Ianine aminotransferase/ALT/ was formerly known as

- a. SGPT
- b. GPT
- c. GGT
- d. GOT
- e. SGOT

Which statement is correct?

- a. Type 2 diabetes is less common than Type 1 diabetes.
- b. Type 1 diabetes patients are older in age
- c. Type 1 diabetic patients are insulin resistant.
- d. DKA is more common in Type 2 diabetes than type 1 diabetes
- e. Type 1 diabetic patients have lean body habitus.

The correct ordering of lipoproteins from the lowest to the highest density is

- a. VLDLs, IDLs, LDLs, chylomicrons
- b. LDLs, IDLs, VLDLs, chylomicrons
- c. LDLs, VLDLs, IDLs, chylomicrons
- d. Chylomicrons, VLDLs, IDLs, LDLs
- e. Chylomicrons, IDLs, VLDLs, LDLs

Choose the correct statement regarding the direct effects of insulin on glycolysis.

- a. It inhibits the conversion of phosphoenolpyruvate into pyruvate
- b. It inhibits the conversion fructose-1,6-bisphosphate into glyceraldehyde-3-phosphate and dihydroxyacetone phosphate
- c. It promotes the conversion of fructose-6-phosphate into fructose-1,6-bisphosphate
- d. It promotes the conversion of glucose-6-phosphate into fructose-6-phosphate
- e. It inhibits the conversion of glucose into glucose-6-phosphate

Introns are cleaved from the exons by the catalytic activity of:

- a. Translocases
- b. Topoisomerases
- c. DNA polymerases
- d. Spliceosomes
- e. RNA polymerases

Enzyme X exhibits maximum activity at pH 6.9. X shows a fairly sharp decrease in its activity when the pH goes much lower than 6.4. One likely interpretation of this pH activity is that:

- a. the enzyme has a metallic cofactor.
- b. a His residue on the enzyme is involved in the reaction.
- c. a Glu residue on the enzyme is involved in the reaction.
- d. the reaction relies on specific acid-base catalysis.
- e. the enzyme is found in gastric secretions.

101 Which molecule is not used during fatty acid synthesis?

- a. Biotin.
- b. Acetyl CoA
- c. NADPH
- d. Bicarbonate.
- e. FADH₂

Why is an increased intake of H₂O recommended for prevention of kidney disease?

- a. Increase the activity of Glyoxylate oxidase
- b. Increased solubilization of calcium oxalate
- c. Increase activity of L-amino acid oxidase
- d. Decreased synthesis of glycine
- e. Decreased the activity of D-amino acid oxidase

In the post-transcriptional modification of proteins polyadenylation at 3' terminal of mRNA is used to:

- a. direct the genes to protein synthesis
- b. be recognized by ribosome for protein synthesis
- c. activate replication process
- d. protect mRNAs from exonuclease degradation
- e. activate translation of genes in ribosome

Which of the following statements is INCORRECT about conjugation reactions?

- a. Conjugation reactions can occur independently
- b. After conjugation, most of the compounds are rendered non-toxic
- c. The reaction can occur after phase-1 reaction
- d. Most of the reactions takes place in the liver
- e. The polarity of the xenobiotics is decreased upon conjugation

105 Which of the following statements about standard solution is correct?

- a. A solution containing little to no analyte of interest
- b. Used to calibrate instrument like spectrophotometer to zero absorbance
- c. A solution containing known concentration of an element or a substance
- d. Used to trace interference comes from either the sample or the reagent
- e. All

After formation of replication fork:

- a. Both the new strands are synthesized discontinuously
- b. One strand is synthesized continuously and the other discontinuously
- c. RNA primer is required only for the synthesis of lagging strand
- d. Supercoiling is resolved by the activity of SSB
- e. All of the above

To calculate the turnover number of an enzyme you need to know the:

- a. initial velocity of the catalyzed reaction at low $[S]$.
- b. initial velocity of the catalyzed reaction at $[S] \gg K_m$.
- c. K_m for the substrate.
- d. enzyme concentration
- e. both B and D

A lipoprotein inversely related to the incidence of coronary atherosclerosis is

- a. IDL
- b. Chylomicrons
- c. LDL
- d. VLDL
- e. HDL

109 During AST test, decrease in absorbance is due to

- a. Reduction of NADH to NAD⁺
- b. Oxidation of NADH to NAD⁺
- c. Oxidation of NAD⁺ to NADH
- d. Reduction of pyruvate to lactate
- e. Reduction of NAD⁺ to NADH

110 Which of the following keeps the DNA termini of chromosomes in the cell from becoming entangled and stick to each other?

- a. Methyl guanosine cap
- b. Telomere
- c. snRNA
- d. Poly A tail
- e. mRNA

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GOOD LUCK NEXT GENERATION!!!